ROB521 Lab 3 Report

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Part 1: turtlebot3_world Localization

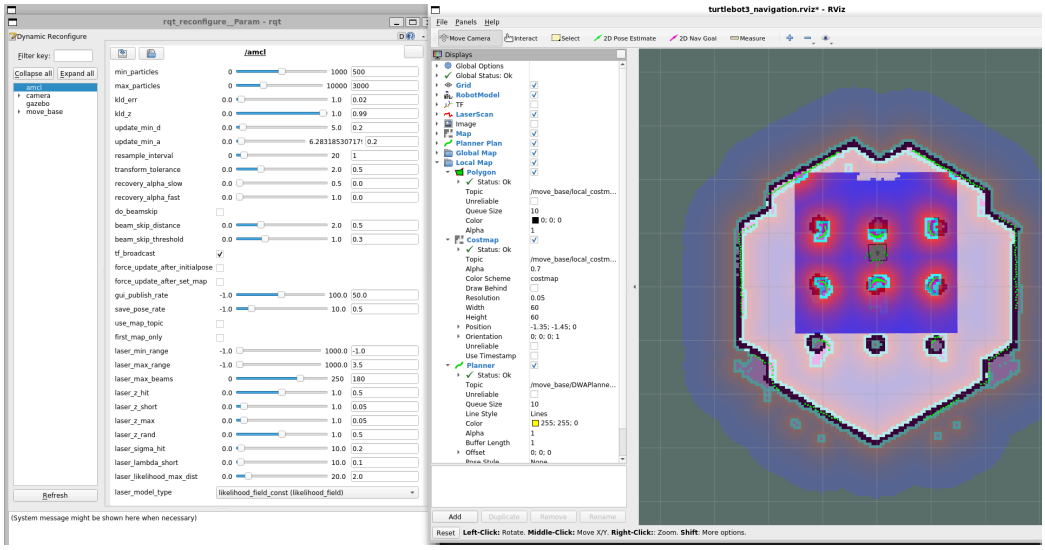

Figure 1: ACML rviz window with converged filter, all default parameter values.

The first set of parameters changed is max and min particles. Both parameters are decreased and the robot is driven around to see resulting localization. Figure 2 shows the localization for min_particles = 6 and max_particles = 68. Three different initial pose estimates were tested for particle filter convergence. These were the obtained values for the minimum number of particles for successful localization, so long as the robot was driven approximately 0.1m/s or lower.

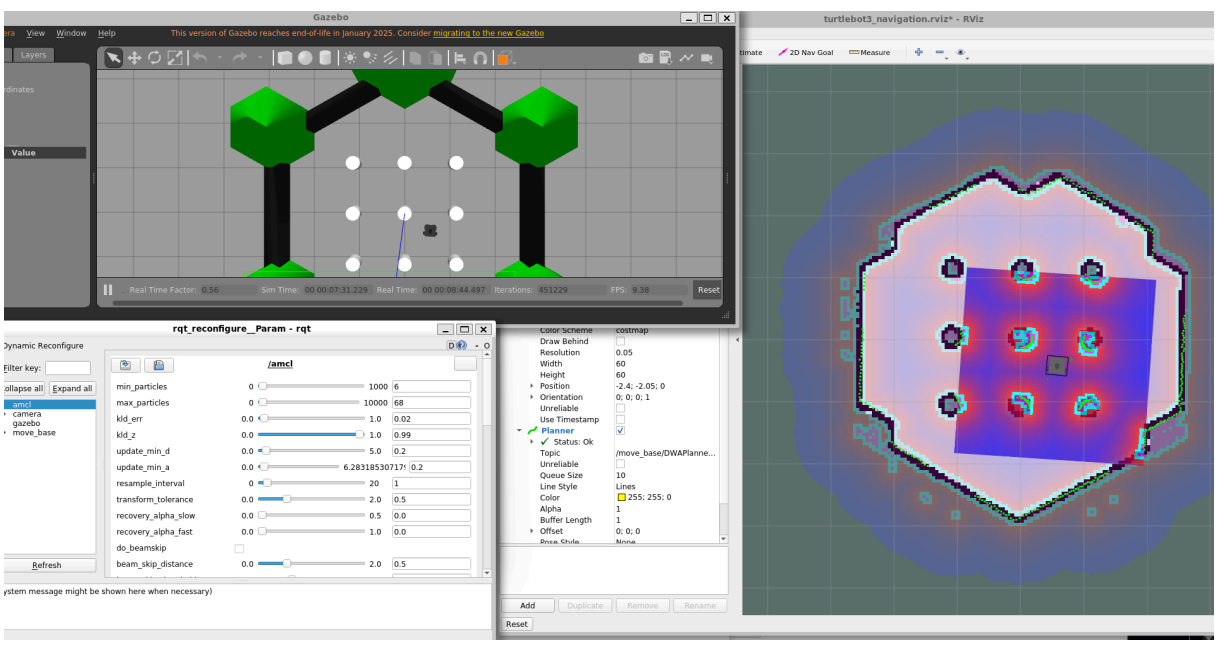

Figure 2: ACML rviz window with converged filter, min_particles = 6, max_particles = 68.

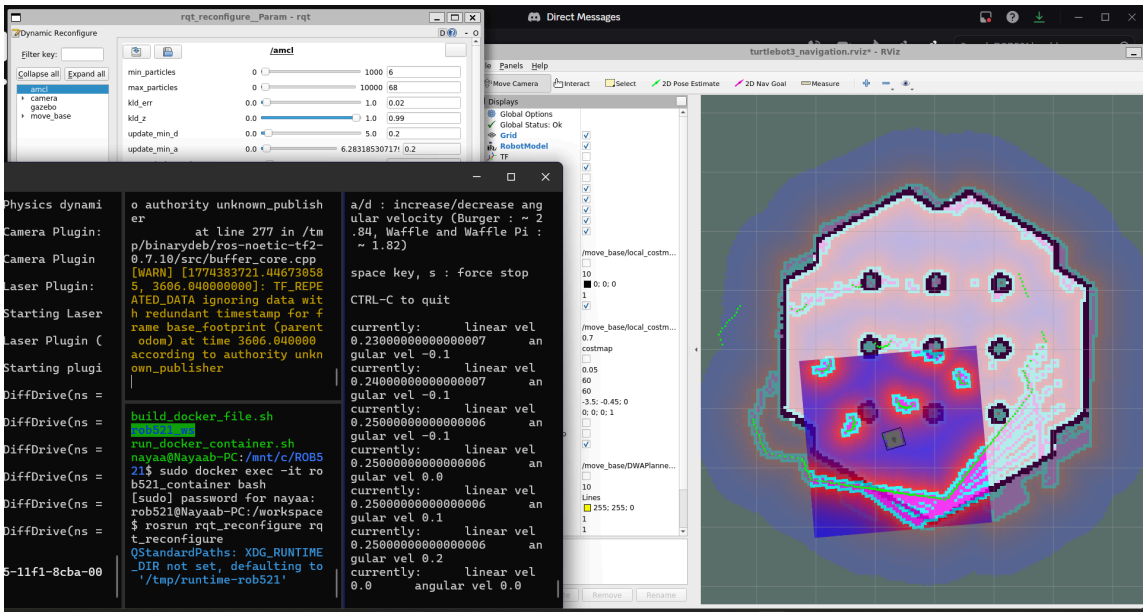

Figure 3: ACML rviz window with minimum number of particles, but velocity of the robot >0.1m/s thus resulting in divergence of localization.

As Figures 4 and 5 display, max and min particle values are returned to default, and odom_alpha values are increased. Past odom_alpha values of around 6.0, there is clear divergence of localization; as such, this is the maximum value observed for filter convergence.

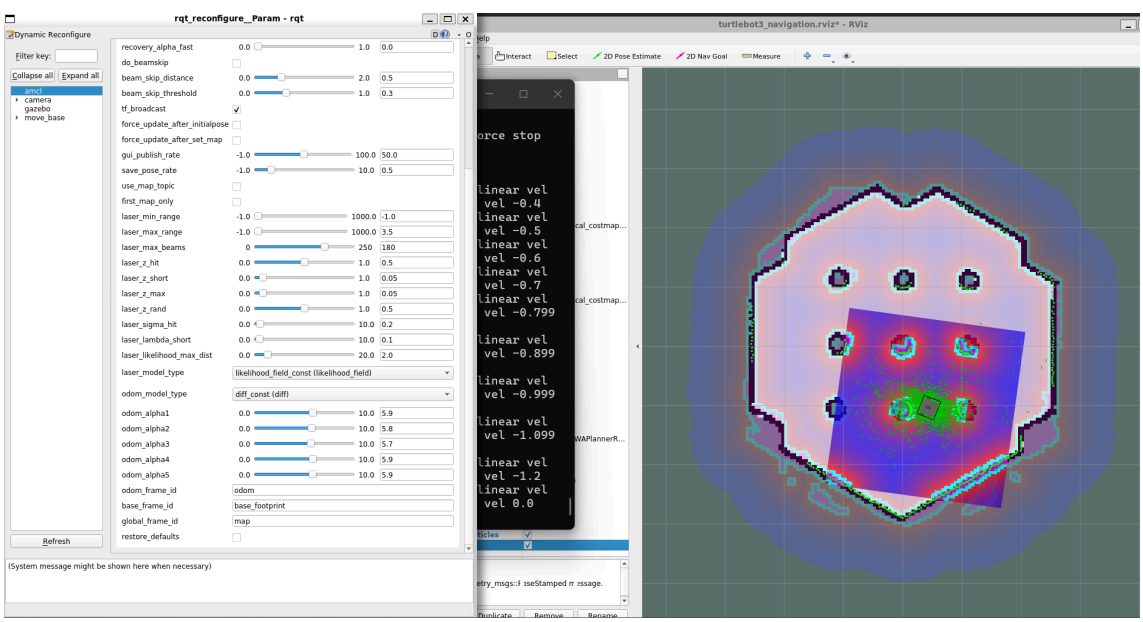

Figure 4: ACML rviz window with converged filter, odom_alpha values = 5.9.

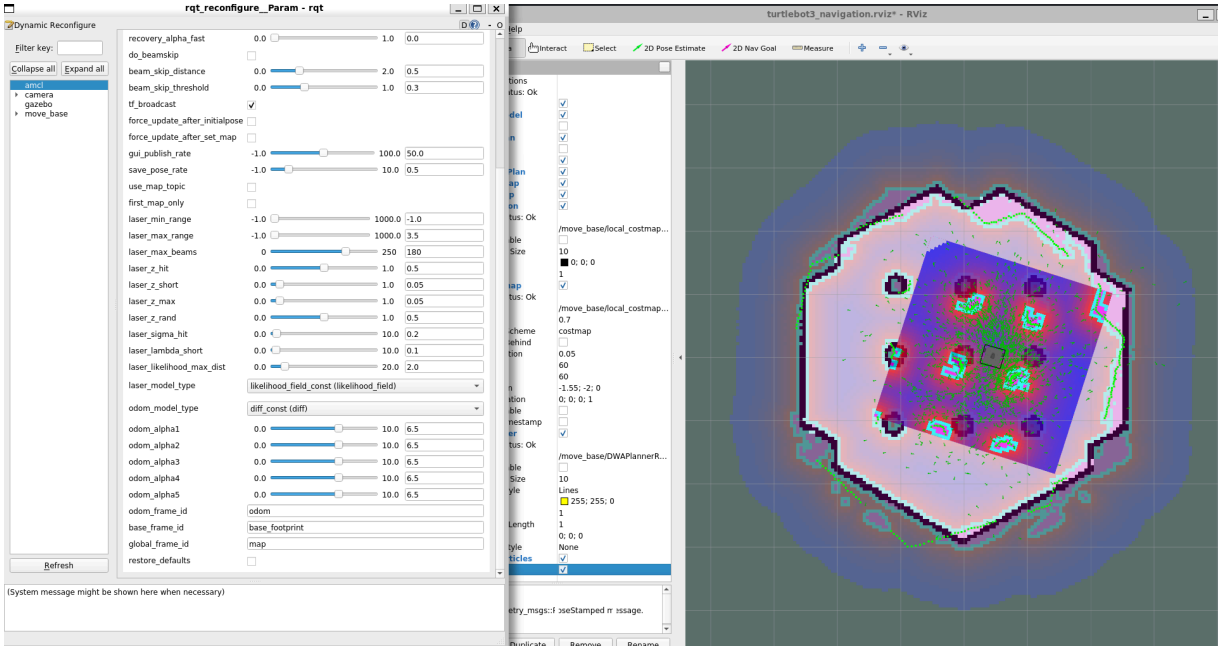

Figure 5: ACML rviz window with odom_alpha values = 6.5, resulting in divergence of localization.

Part 2: turtlebot3_world Mapping

The map generated using Gmapping shows a clear improvement over the map produced in Lab 3 due to the inclusion of a localization component based on a particle filter. In Lab 3, mapping relied primarily on wheel odometry, which accumulates error over time and results in drift. This typically causes walls to appear misaligned, duplicated, or blurred.

In contrast, Gmapping simultaneously performs localization and mapping using a particle filter combined with laser scan matching. The localization component continuously estimates the robot's pose by comparing incoming LiDAR scans with the existing map, allowing it to correct for odometry drift. As a result, the generated map is more consistent and geometrically accurate.

This improvement is evident in the provided map, where the walls are well-defined and form a coherent boundary, and the internal obstacles are clearly and consistently placed. The structure appears symmetric and aligned, indicating that the robot's pose estimate remained accurate throughout the mapping process. Additionally, loop closure allows Gmapping to correct accumulated errors when revisiting previously mapped areas, further improving global consistency.

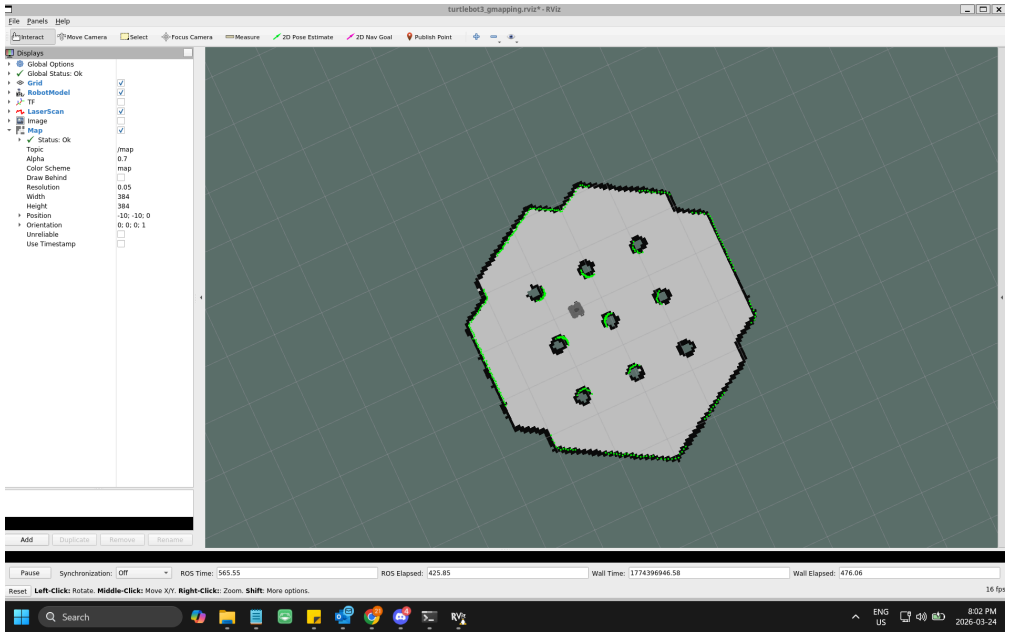

Figure 6: Results of mapping `turtlebot3_world` using Gmapping

Part 3: willowgarage_world Mapping

Comparison with Lab 2 map

The generated map is generally consistent with the provided Lab 2 map in terms of overall layout and structure. Major features such as corridors and open spaces are correctly captured, and most walls are well-aligned. However, the major difference appears inside the large room at the end of the outlined purple path. Since the mapping was not perfect, drift accumulated as the robot was mapping the wall such that when it revisited an older arena, the newly mapped area collided with the older mapped area.

Mapping issues encountered

A key issue observed in the generated map occurs along a long straight corridor that leads into a loop closure. At the point where the robot returns to the starting area, the walls do not align properly and instead overlap or “collide” with each other. This issue is primarily caused by accumulated odometry drift over long distances. As the robot travels through the corridor without revisiting known areas, small pose estimation errors build up. Although Gmapping uses scan matching to correct these errors, it becomes less effective in long, feature-sparse environments such as straight walls.

Loop closure issues

When attempting to close the loop, the map does not perfectly align with the previously mapped region. Instead, the walls at the meeting point intersect or misalign, showing that the loop

closure was imperfect. This occurs because the particle filter's estimate of the robot's pose has diverged from the true pose, and the correction applied during loop closure is insufficient to fully resolve the accumulated error. This is especially prevalent after long straight walls, which can be seen at the end of the purple path.

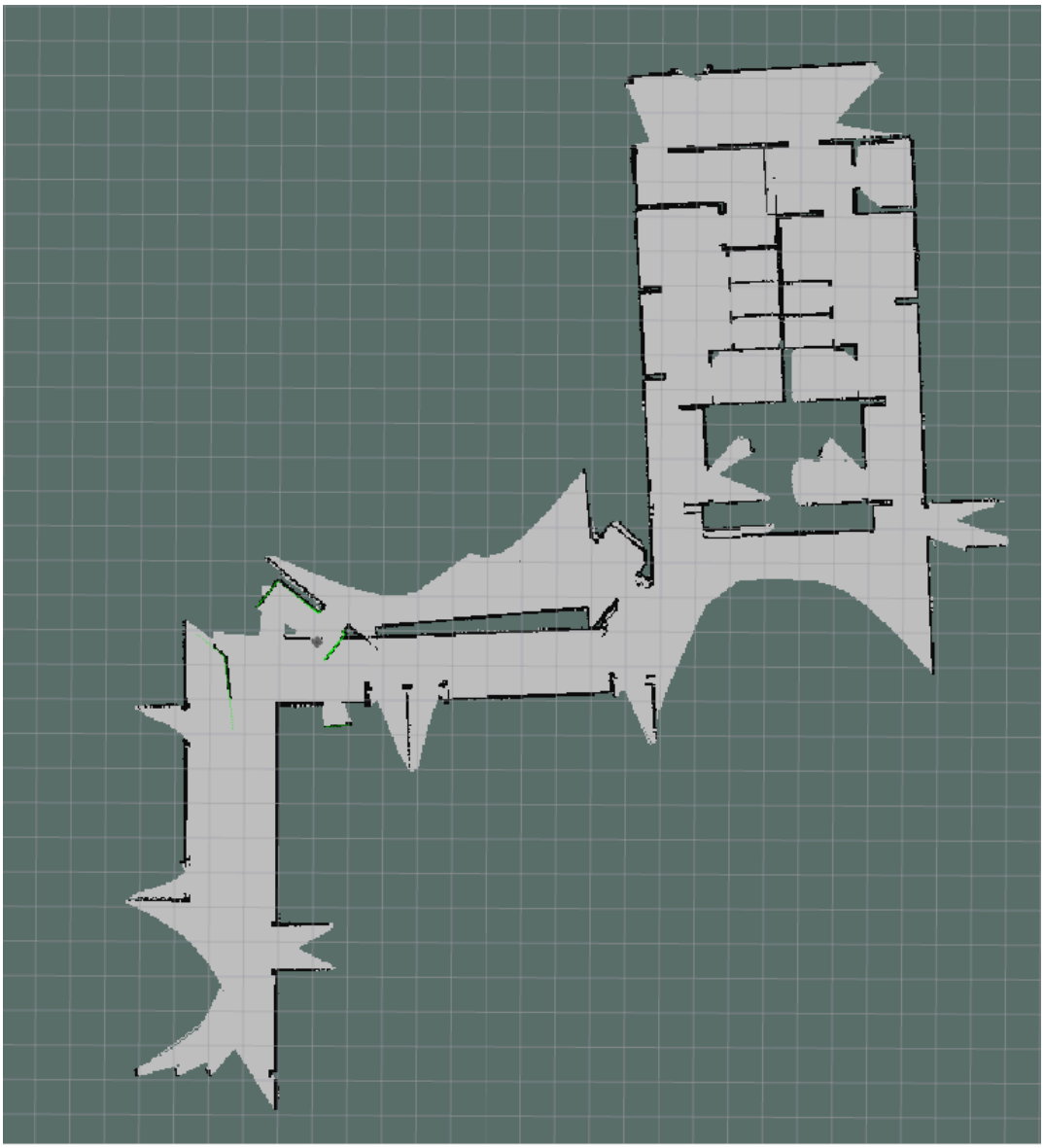

Figure 7: Map of `willowgarage_world` generated using Gmapping as part of Lab4

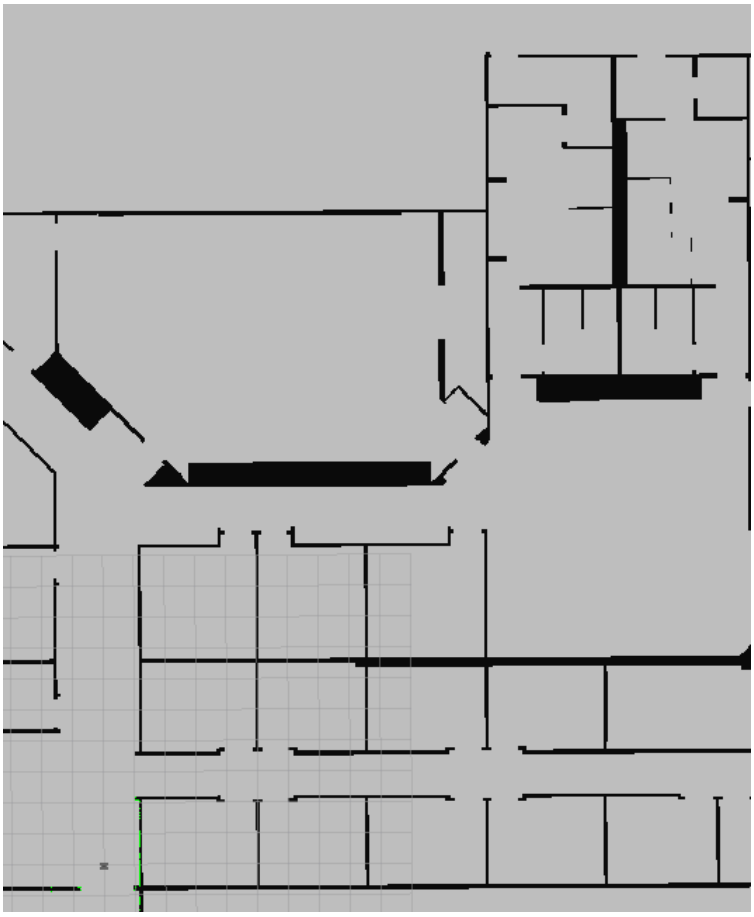

Figure 8: Complete map of `willowgarage_world` provided in Lab2

Part 4: Myhal Environment Mapping

Mapping Issues Encountered & Real-life Versus Simulation Comparison

The primary mapping issue visible in the Myhal scan is map blurring and jumping, where wall boundaries appear fuzzy or jumping further away quickly. This is caused by noisy wheel odometry and the limitations of the mapping. When the robot's physical movement deviates from its internal motion model due to floor friction or wheel slippage, the particle filter may fail to perfectly align new lidar scans with the existing map. Other factors affecting mapping include the real-life open arena where the lidar could encounter surfaces with varying reflective properties as a result of the material or light interference from the windows in the room, which can cause the sensor to return inconsistent distance measurements. The transition from simulation to a real-life deployment confirms that real-world hardware is subject to physical errors that do not exist in a perfect simulation. These factors make maintaining a consistent trajectory much more difficult in person, often requiring more precise parameter tuning to prevent the localization and mapping from diverging.

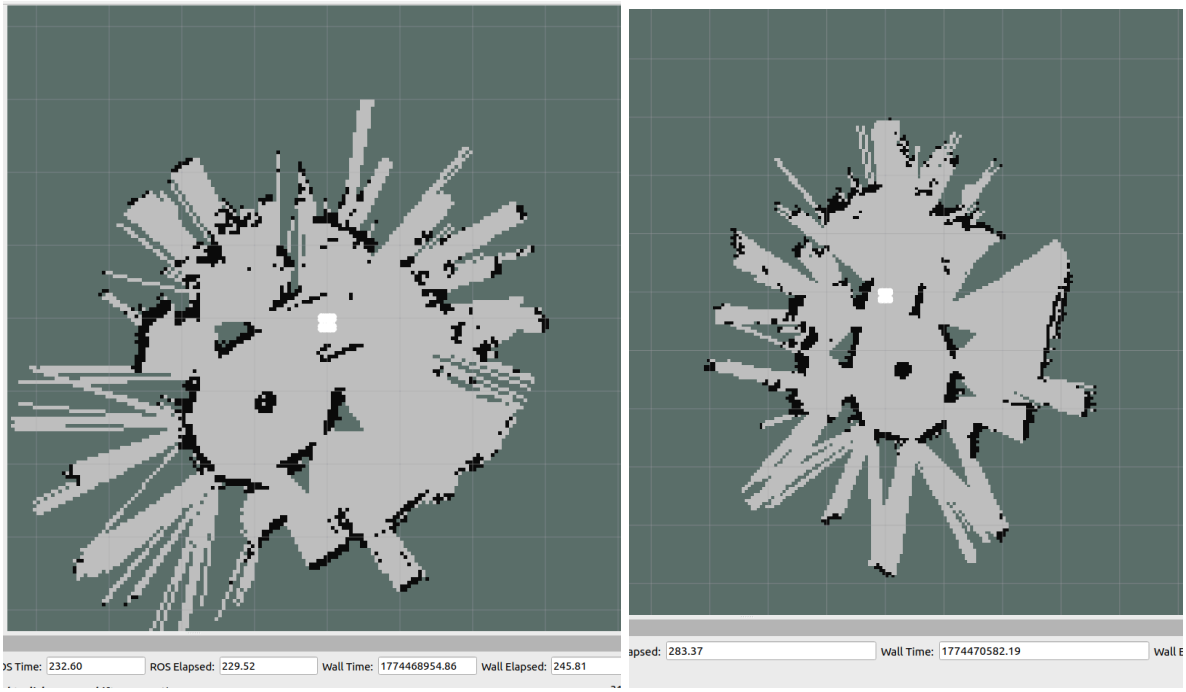

Figure 9: Complete map of Myhal Testing Environment (2 separate trials)